

WASP-2b

R-0267

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_240708 · created 2026-07-17T10:02:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460499.8398 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.013
Transit depth (Rp/Rs)^2	2.01 +/- 0.38 [%]
Orbital Inclination (inc)	85.03 +/- 1.34
Airmass coefficient 1 (a1)	0.978 +/- 0.012
Airmass coefficient 2 (a2)	0.018 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	#3 - [491, 120]
Optimal Method	PSF photometry
Transit Duration (day)	0.074 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.142 ± 0.013 vs 0.1326 (deviation 0.55σ)
Duration fit vs published	0.074 d vs 0.0737 d (deviation 0.01σ)
Mid-time O–C	7.5 min
Depth SNR	8.3
Residual scatter	1.27 %

Light curve

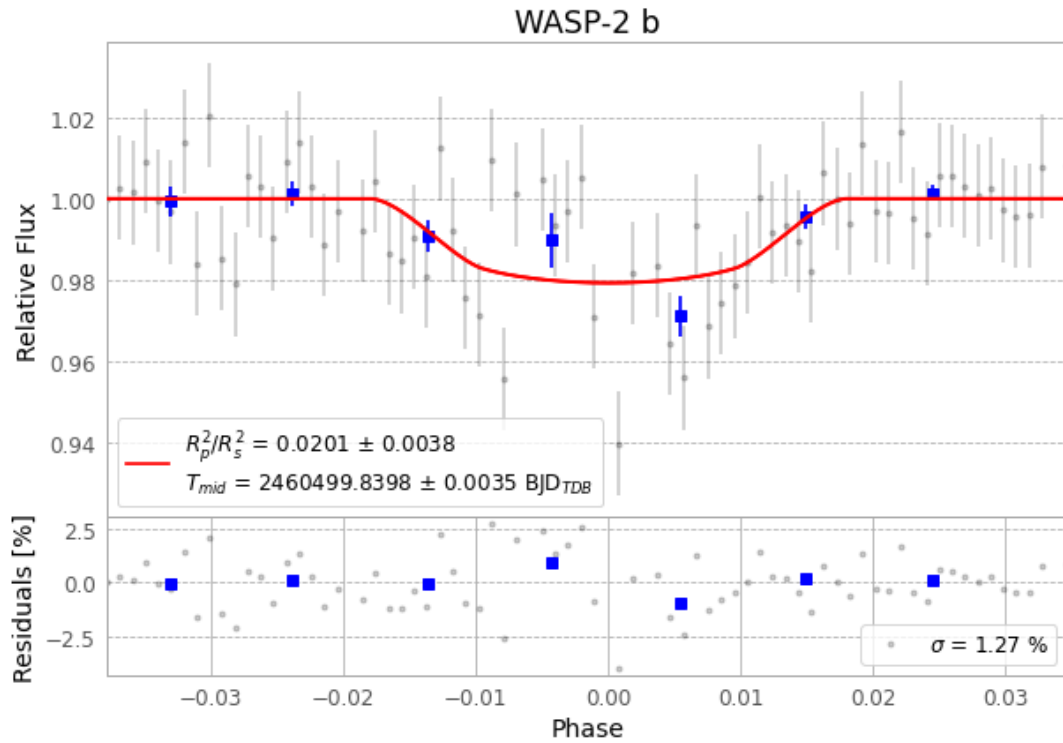


Figure 1 — FinalLightCurve_WASP-2 b_2024-07-07.png (SHA-256 c96f28167c23d2af...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a8052350125dbdf6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 7dbc7ef

Data provenance

75 FITS frames (49 MB), WASP-2240708060315.FITS → WASP-2240708094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-240708.log (92e60ab20977...), FinalLightCurve_WASP-2 b_2024-07-07.png (c96f28167c23...), FinalLightCurve_WASP-2 b_2024-07-07.csv (0db48bb88dfc...)

Compute resources

Wall time 145.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 10:02Z.